

Smart City Traffic Intelligence

An End-to-End Data Analytics, Machine Learning and MLOps Study

NUS School of Computing - AI, Machine Learning and Data Science Capstone Project

Metro Interstate Traffic Volume | Westbound I-94, Minneapolis-St Paul

Final integrated report | Parts 1-3

Project scope
48,204 source records from 2 October 2012 to 30 September 2018, consolidated to 40,575 unique hourly observations. The project progresses from descriptive analytics and data-quality assessment to a reproducible ML pipeline, predictive modelling, explainability, MLOps, monitoring and Responsible AI evaluation.

Executive Summary

This capstone followed the traffic data from its raw form through to a working machine-learning and MLOps workflow. Historical westbound I-94 traffic and weather records near Minneapolis-St Paul were first examined for patterns and data-quality problems. The cleaned data were then prepared in Python for modelling, before being used for supervised and unsupervised learning, model explanation, travel-timing recommendations, MLflow tracking, deployment and monitoring. Responsible AI considerations were assessed alongside these technical stages rather than treated as a separate issue at the end.

The clearest finding was the importance of time. In Part 1, hour of day alone explained 78.2% of the variation in traffic volume, whereas temperature had only a weak overall relationship with traffic. Once observations were compared within similar time periods, several apparent weather differences became much smaller. This shaped the features created in Part 2. It was also reflected later in Part 3, where the cyclical hour variables were the most influential Random Forest features in the SHAP analysis.

For the supervised models in Part 3, Random Forest gave the strongest results. On the chronological test set, the Random Forest regressor had an MAE of 249.16 vehicles and an R^2 of 0.9567, compared with 720.19 and 0.7634 for Linear Regression. The feed-forward neural network fell between these two models. The high-risk proxy classifier also produced strong metrics, but these need to be interpreted carefully: the target was deliberately constructed for the assignment, so the result shows how well the model learned that proxy rather than how well it would predict real accidents.

The final stage moved beyond simply training models. MLflow was used to keep track of experiments and model versions, FastAPI was used to demonstrate model serving, and a simple monitoring process was built around changes in prediction error. The project therefore shows the full path from raw data to an interpretable and monitored model, while still being limited by the age and coverage of the dataset, its focus on one corridor, possible drift by 2026, and the absence of real accident labels.

1. Project Context, Data Quality and Data Foundation

The source data contained 48,204 records for westbound I-94 between 2 October 2012 and 30 September 2018. Before analysing traffic patterns, it was important to establish what these rows actually represented. The 48,204 records were not 48,204 separate traffic hours. Some timestamps appeared more than once because more than one weather condition had been recorded for the same hour. Treating every row as a separate traffic observation would therefore have counted some hours more than once, so the later pipeline consolidated the data to one observation per hour.

The data also contained problems that could not all be handled in the same way. There was a 308-day gap from 8 August 2014 to 11 June 2015 and 11,976 missing hours overall; these missing observations could not be recovered and were treated as a limitation. Other problems could be corrected more directly. Ten temperature readings were recorded as 0 K, one hourly rainfall value was 9,831.3 mm, and the holiday field marked a holiday only at 00:00 rather than throughout the day. Annual coverage also varied substantially. These issues became the starting point for the cleaning and feature-engineering work in Part 2.

Data-quality issue	Evidence identified	Treatment / implication
Exact duplicate records	17 exact duplicates identified during pipeline cleaning	Removed before further processing

Repeated hourly timestamps	48,204 raw rows represented 40,575 unique hours	Consolidated to one hourly observation
Missing historical coverage	308-day gap; 11,976 missing hours overall	Documented as a limitation; not imputed
Invalid temperature values	10 observations recorded 0 K	Identified as implausible and imputed
Implausible rainfall	One observation recorded 9,831.3 mm	Identified as implausible and imputed
Incomplete holiday representation	Holiday field marked only 00:00	Reconstructed as a whole-day indicator
Unequal annual coverage	Coverage ranged from 23.9% in 2012 to 99.5% in 2017	Avoided direct comparison of raw annual totals

1.1 Traffic Patterns and Analytical Findings

SQLite, statistical analysis and Power BI were used in Part 1 to understand the traffic patterns before any machine-learning models were built. Time of day stood out immediately. Hour alone explained 78.2% of the variation in traffic volume. In 2017, average traffic was only about 385.8 vehicles at 03:00, rising to 4,819.5 at 07:00 and 5,834.4 at 16:00. This was not just a descriptive result. It influenced the later modelling choices: Part 2 represented hour as a cyclical feature, and those hour variables subsequently became the two most influential features in the Random Forest SHAP analysis.

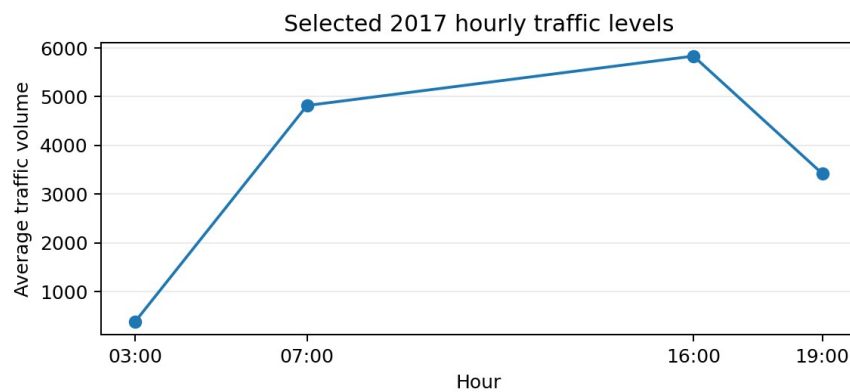


Figure 1. Selected points from the 2017 hourly traffic profile.

Weather initially appeared to be important, but much of that relationship weakened when time was taken into account. Temperature and traffic volume had an overall correlation of only $r = 0.139$, and this fell to $r = 0.035$ when observations were compared within the same month and hour. A similar pattern appeared when clear and cloudy conditions were compared: the apparent difference in congestion odds was largely lost after controlling for hour of day. This suggests that some of the raw weather differences were actually reflecting when those conditions happened to occur rather than a strong independent weather effect.

Day type provided another useful signal. Across 2015-2017, Labor Day traffic was only about 65-68% of that on an ordinary September Monday. This supported adding weekend and whole-day holiday indicators to the later dataset. Year-to-year totals, however, were less straightforward because the amount of recorded data differed substantially between years. For that reason, raw annual totals were not treated as directly comparable; well-covered periods and traffic per recorded hour gave a fairer basis for comparison.

2. Reproducible Data Pipeline and Feature Engineering

Part 2 turned the problems found in Part 1 into a repeatable Python cleaning process. The pipeline started with 48,204 records and removed 17 exact duplicates. It also identified the 10 invalid temperature values and the single implausible rainfall value for imputation. Repeated timestamps were handled separately because they were not necessarily duplicate records: in many cases they represented different weather descriptions attached to the same traffic hour.

After cleaning, the data were consolidated into 40,575 unique hourly observations and expanded to 36 variables. The new variables captured hour, day of week, month, year, weekends, holidays and weather, together with standardized versions of continuous measurements. Hour and day of week were also encoded with sine and cosine functions. This matters because time is cyclical: 23:00 and 00:00 are neighbouring hours even though their ordinary numerical values are far apart. The same principle applies to the end and beginning of a week.

Pipeline stage	Output / purpose
Raw source	48,204 rows
Validation and cleaning	48,187 rows after 17 exact duplicates removed
Hourly consolidation	40,575 unique hourly observations
Feature engineering	36 variables
ML-ready dataset	Prepared dataset used for Part 3 modelling

Congestion was divided into four levels using the traffic-volume quartiles rather than an assumed road-capacity threshold. Low represented $\leq 1,248.5$ vehicles, Medium $> 1,248.5$ -3,427, High $> 3,427$ -4,952 and Severe $> 4,952$. Because the cut-offs came from quartiles, the four groups were almost the same size: 10,144 Low, 10,144 Medium, 10,145 High and 10,142 Severe observations. These categories were later reused when the high-risk proxy for Part 3 was constructed.

The cleaned dataset still showed the same broad traffic pattern seen in Part 1, which provided a useful check that the transformation process had not removed the underlying signal. Weekday traffic peaked at about 6,241 vehicles per hour at 16:00, while the weekend maximum was lower, at about 4,408 vehicles at 13:00. The workflow also saved logs and figures so that the cleaning steps could be reviewed later. A small command-line application made the processed data usable for date and time queries, high-traffic searches, weekday/weekend comparisons and historical travel-time recommendations. This provided a simple bridge between the earlier analysis and the API developed in Part 3.

3. Machine Learning, AI and MLOps

For Part 3, 24 predictors were selected from the engineered dataset. They covered time, day type, holidays, month and year, temperature, rain, snow, cloud cover and individual weather conditions. Instead of randomly mixing observations from different years, the data were split in time order. The earlier 80% (32,460 observations) was used for training and the most recent 20% (8,115 observations) was kept for testing. This made the test set a later period than the data seen during training.

The dataset did not contain actual accident outcomes, so the classification task required a proxy target. An observation was labelled high risk when High or Severe congestion occurred together with adverse weather. This produced 5,605 high-risk proxy observations, or 13.81% of the dataset. Traffic volume and the derived congestion category were deliberately left out of the classifier inputs because they had already been used to define the target. Even with this precaution, the target

remains an artificial label. The classification results therefore show how well the models learned the proxy pattern, not how accurately they would predict real accidents.

3.1 Task 1 - Supervised Machine Learning

3.1.1 High-Risk Proxy Classification

Logistic Regression and Random Forest were tested on the same later time period. Random Forest gave the stronger overall set of results, although Logistic Regression still achieved very high recall.

Classification model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.9506	0.7658	0.9875	0.8626	0.9923
Random Forest	0.9832	0.9234	0.9741	0.9481	0.9976

The high values in this table should therefore be read in the context of the constructed target. They show that the proxy pattern was learnable from the available features; they do not show that the models can predict real-world traffic accidents.

3.1.2 Traffic-Volume Regression

Regression model	MAE (vehicles)	R ²
Linear Regression	720.19	0.7634
Random Forest	249.16	0.9567

For traffic-volume prediction, the difference between the two models was much clearer. Random Forest reduced the average absolute error from 720.19 to 249.16 vehicles per hour. Its R² of 0.9567 means that it accounted for about 95.7% of the variation in traffic volume in the held-out test period. This is a measure of explained variation, not a statement that the model was '95.7% accurate'.

3.2 Task 2 - Unsupervised Learning and Association Rules

K-means clustering was used to see whether the traffic observations naturally separated into recurring combinations of traffic volume and weather. Solutions from $k = 2$ to $k = 6$ were compared using the silhouette score, which gives an indication of how well observations fit their own cluster compared with neighbouring clusters. Of the values tested, $k = 6$ had the highest score (0.3433). The score was only moderate, so the six clusters were treated as useful descriptive groupings rather than sharply distinct traffic states.

The resulting clusters included both low- and high-volume conditions under different weather patterns, as well as a small cluster associated with heavy rain. Association-rule mining produced a particularly clear result: Weekend + Night $\rightarrow$ Low congestion. The rule appeared in 6.44% of all observations (support), and 89.63% of weekend-night observations were Low congestion (confidence). Its lift of 3.585 means that Low congestion was about 3.6 times more common under this combination than it was in the dataset overall. The rule describes an association and should not be interpreted as showing that weekends or nighttime cause low congestion.

3.3 Task 3 - Neural Network and Model Explainability

The feed-forward neural network was trained on the same traffic-volume prediction problem. Its MAE was 512.83 and its R² was 0.8653. This was a clear improvement over Linear Regression, but it was still well behind Random Forest. In this dataset, using a neural network simply because it was a more complex model did not lead to better predictions.

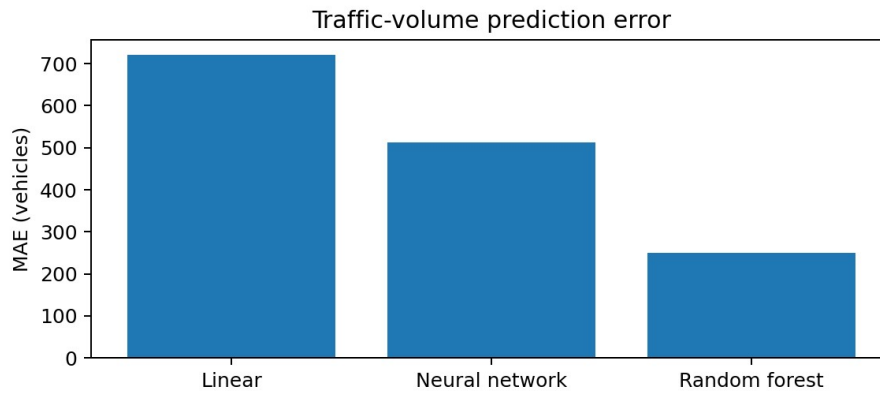


Figure 2. Test MAE for the three traffic-volume prediction approaches. Lower values indicate better performance.

To understand what the strongest regression model was relying on, SHAP was applied to the Random Forest regressor. The five most influential features by mean absolute SHAP value were `hour_cos`, `hour_sin`, `day_of_week_sin`, `is_weekend` and `day_of_week_cos`. This was a useful link back to the earlier analysis: the temporal pattern first seen in Part 1, and deliberately encoded in Part 2, was also what the model relied on most heavily in Part 3. The chart ranks the size of each feature's influence; it does not by itself show whether a feature pushes traffic predictions up or down.

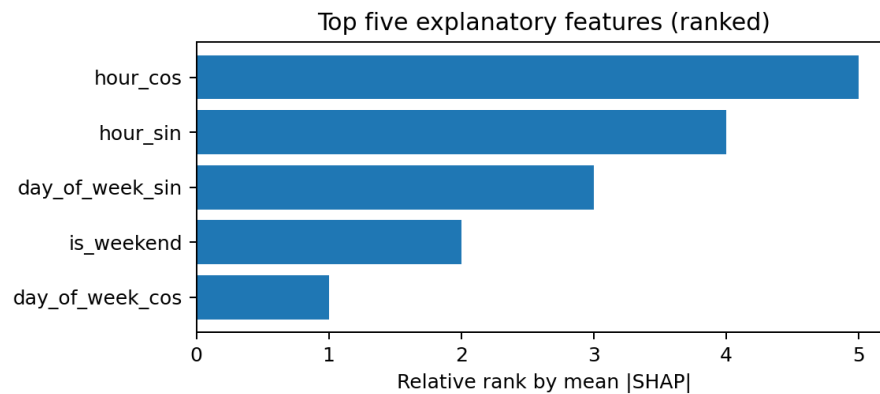


Figure 3. Top five Random Forest features ranked by mean absolute SHAP magnitude.

3.4 Task 4 - Advanced AI Technique: MLflow

MLflow was used to keep the modelling work traceable rather than leaving each model as an isolated output file. A local SQLite backend was specified explicitly so that experiment details and model-registry versions were stored in the same known location and remained available when MLflow was restarted. Five main model runs were recorded in this local experiment-tracking setup.

Registry version	Model	Role	MAE	R ²
Version 1	Linear Regression	Baseline	720.19	0.7634
Version 2	Random Forest Regression	Improved candidate	249.16	0.9567

The already fitted models were then added to the registry without retraining them. Linear Regression was kept as Version 1 and Random Forest Regression as Version 2, so the improvement could be seen without losing the original baseline. This was a local capstone implementation rather than a shared production MLflow server, but it demonstrated how model history can be kept persistent and reviewable.

3.5 Task 5 - Traffic Recommendation System

The dataset covers only one corridor, so there was no basis for recommending alternative routes. The recommendation component therefore focused on identifying the times when this corridor was historically quieter. Traffic was summarized by hour, day type and weather. The lowest-volume periods generally fell between 02:00 and 04:00. These are useful as historical minima, although they are clearly not practical travel times for many commuters.

Condition	Lowest historical period	Mean traffic
Weekday - normal weather	02:00	305
Weekday - low visibility	02:00	300
Weekday - severe weather	02:00	301
Weekend - normal weather	04:00	384
Weekend - low visibility	03:00	402
Weekend - severe weather	04:00	355

A more useful interpretation is therefore to consider these minima alongside realistic off-peak periods. For example, Part 1 showed that average traffic at 19:00 in 2017 was about 41% lower than at the 16:00 peak. The recommendation output is best understood as historical timing guidance for this particular corridor, not as live navigation or route advice.

3.6 Task 6 - MLOps and Deployment Simulation

3.6.1 Model Versioning and Experiment Tracking

The MLflow registry kept Linear Regression as Version 1 and Random Forest Regression as Version 2. Using the SQLite backend meant that these records, their metrics and their version history remained available across sessions rather than depending on a temporary run of the interface.

3.6.2 Deployment

FastAPI was then used to show how a trained model could be exposed outside the training script. The /predict endpoint accepted the 24 model inputs and returned the proxy class together with its probability. The response also included a warning that the output referred to the constructed high-risk proxy and not to actual accident risk.

3.6.3 Monitoring and Alerting

A simple monitoring check was demonstrated by dividing the held-out period into an earlier reference window and a later monitoring window. Linear Regression MAE was 723.72 vehicles in the reference period and 716.66 in the later period, a change of -0.98%. The alert rule was set to trigger if MAE worsened by more than 20%, so this run returned PASS - Normal. This only means that the chosen error threshold was not breached. It does not show that every input feature remained stable or rule out other forms of drift.

MLOps element	Implementation	Result / value
Experiment tracking	MLflow + SQLite	Model runs, parameters and metrics retained
Model versioning	MLflow Registry	V1 Linear -> V2 Random Forest
Serving	FastAPI	Inputs returned a proxy prediction
Monitoring	Chronological MAE comparison	-0.98% change; PASS / Normal

3.7 Task 7 - Responsible and Sustainable AI

Responsible AI was considered in terms that were relevant to the data available. The dataset contains no demographic attributes, so a conventional demographic fairness comparison was not possible. Instead, the main concerns were where and when the data were collected, whether model errors changed under different operating conditions, and the fact that the high-risk target was constructed rather than observed. These limitations matter because they restrict how confidently the outputs could be used in real traffic decisions.

The error analysis showed why a single overall performance number can be misleading. Linear Regression MAE rose from 689.69 vehicles in normal weather to 781.80 in adverse weather, about 13.4% higher. The difference by time of day was larger still: morning MAE was 1,059.78 compared with 491.47 in the afternoon. A model could therefore appear acceptable overall while performing noticeably worse under particular conditions, which is a strong reason to monitor those conditions separately.

If the system were taken beyond this capstone, model outputs would still need human review and validation on current data, especially where the high-risk proxy was involved. There was also a practical resource trade-off between models. Random Forest gave the best regression results, but its saved model was much larger than the linear and neural-network models. Lossless joblib compression reduced the Random Forest artifact from about 281 MB to 52 MB without changing predictions in the verification sample. This reduced storage requirements, although it did not change the model itself through quantisation or pruning.

4. Integrated Findings and Mobility Implications

The same broad result appeared repeatedly as the project developed: time mattered more than weather for explaining traffic volume on this corridor. It first appeared in Part 1, where hour of day explained a large share of the variation and several weather relationships weakened after time was taken into account. Part 2 then encoded this temporal structure directly into the model inputs. Finally, the SHAP results in Part 3 showed that the cyclical hour variables were the most influential Random Forest features. Seeing the same pattern in the exploratory analysis, the engineered features and the model explanation gives greater confidence that it is a genuine feature of this dataset rather than an artefact of one analytical method.

This does not mean that weather was irrelevant. Weather still helped describe operating conditions, formed part of the high-risk proxy, and appeared in the clustering analysis. Model error was also about 13.4% higher in adverse weather than in normal weather. The more balanced interpretation is therefore that weather added useful context, but it was not the main factor explaining how much traffic was present.

The project also made clear that model quality depended on the data preparation that came before it. Repeated timestamps had to be consolidated, holidays had to be represented across the whole day, and implausible sensor readings had to be handled before modelling. Once these issues were addressed, Random Forest produced the strongest traffic-volume predictions on the later test period. The same temporal patterns could then be translated into simple historical travel-timing guidance.

Good test-set performance was only one part of the final workflow. A model would also need to be identifiable, versioned, served and monitored if it were to be used beyond the project. MLflow and SQLite kept the experiment and model history persistent, FastAPI demonstrated how predictions could be requested through an API, and the monitoring script checked whether prediction error had

deteriorated. The subgroup analysis also showed why monitoring only an overall metric would be insufficient. Together, these steps moved the project from model training towards a small but auditable decision-support workflow.

5. Limitations and Conclusion

5.1 Limitations

Several limitations restrict how far the findings can be generalised. The data cover only the westbound direction of one corridor, so they cannot describe route choice or traffic displacement across a wider road network. Missing periods and unequal annual coverage also make long-term comparisons less certain. Because the data are observational, the relationships reported here should be read as associations rather than causal effects. In addition, the classification target was a constructed high-risk proxy rather than a record of actual accidents, and the recommendation system used historical traffic rather than live conditions or alternative-route data.

The age of the data is another important limitation. The final observations are from September 2018, while this project is being completed in 2026. Traffic patterns may have changed since then because of changes in working habits, road infrastructure, surrounding development, public transport use and other factors. In machine-learning terms, the distribution of the inputs may have shifted (data drift), and the relationship between those inputs and traffic volume may also have changed (concept drift). The PASS result from Task 6 only compared two periods within the historical dataset; it cannot show that the model is still valid in 2026. Current traffic data would be needed before using the model operationally, and retraining might be required.

5.2 Conclusion

Taken together, the three parts form a single workflow rather than separate exercises. Part 1 established what could and could not be trusted in the raw data. Part 2 converted those findings into a reproducible hourly dataset with features suitable for modelling. Part 3 then used that dataset for prediction, clustering, explainability, recommendation, experiment tracking, deployment and monitoring.

The most consistent finding was the importance of recurring time patterns. Random Forest gave the best traffic-volume predictions of the models tested, and its SHAP explanation again highlighted hour and day features. At the same time, the project exposed several reasons not to judge a system by headline model metrics alone: the historical data are incomplete and now several years old, prediction error changes with operating conditions, and the risk classifier is based on a proxy rather than real accident outcomes. The final result is therefore best viewed as a reproducible demonstration of how raw traffic data can be turned into an interpretable and monitored decision-support workflow, rather than as a production traffic-management system.

Appendix A - Key Reproducibility Artifacts

Area	Primary artifact / evidence
Part 1	SQLite/SQL analysis, Power BI dashboard, Data Analytics Insights Report
Part 2	pipeline.py, feature_engineering.py, visualisations, CLI, pipeline.log, Part 2 report
Part 3 supervised	task1_supervised.py, saved model artifacts and result tables
Unsupervised	task2_unsupervised.py, cluster profiles and association-

	rule outputs
Neural network / SHAP	task3_neural_network.py and explainability outputs
MLflow	mlflow.db, experiment records, model registry
Deployment	FastAPI application and prediction endpoint
Recommendation	Historical travel-timing recommendation outputs
Monitoring	Chronological error-monitoring output
Responsible AI	Bias/fairness analysis and sustainability discussion
Version control	README documentation and Git commit history